

SHAWINIGAN, QC, Canada

WMO: 713700

Lat: 46.564N

Lon: 72.733W

Elev: 361

StdP: 14.51

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-9.2	-24.0	1.5	-11.8	-19.4	1.9	-7.9	19.2	16.4	17.2	17.3	4.7	320	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	85.1	70.2	81.9	68.0	79.0	66.6	73.1	80.7	71.0	78.1	69.2	76.1	9.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.6	114.0	77.6	68.6	106.6	75.0	66.6	99.3	73.1	37.0	80.7	35.1	78.0	33.6	76.2	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.3	12.8	-19.7	90.4	4.7	3.6	-23.1	93.0	-25.9	95.2	-28.6	97.2	-32.0	99.8		
(5)			-20.0	77.0	4.5	2.5	-23.2	78.8	-25.9	80.3	-28.4	81.7	-31.7	83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	11.4	14.9	25.8	39.9	53.9	62.7	67.8	65.9	58.2	45.3	32.9	18.8		
	DBStd	21.47	12.43	10.99	10.59	7.96	7.45	6.62	5.19	5.24	7.15	7.36	8.50	10.95		
	HDD50	4963	1198	982	751	315	40	1	0	0	11	184	515	966		
	HDD65	8823	1663	1402	1216	754	353	120	29	53	227	611	964	1431		
	CDD50	1898	0	0	0	11	162	383	552	494	256	39	1	0		
	CDD65	282	0	0	0	0	10	52	116	82	22	0	0	0		
	CDH74	2181	0	0	0	7	157	462	864	532	157	2	0	0		
	CDH80	498	0	0	0	0	38	121	211	93	35	0	0	0		
	WSAvg	6.0	5.9	6.3	6.8	7.0	6.5	5.8	5.4	5.4	5.4	6.0	6.0	6.0		
	PrecAvg	43.7	3.0	2.8	2.9	3.3	3.7	3.9	3.9	4.1	3.9	3.9	3.8	4.4		
(15) Precipitation	PrecMax	55.7	4.8	8.2	6.0	9.6	10.5	9.9	7.3	8.8	7.1	10.9	7.2	7.5		
	PrecMin	34.8	0.9	0.2	0.5	0.5	1.1	1.9	0.4	1.2	0.4	0.8	1.4	1.1		
	PrecStd	5.6	1.0	1.9	1.2	1.9	2.0	1.6	1.5	1.8	1.4	2.0	1.5	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	40.8	54.1	70.9	84.9	88.1	89.8	86.4	85.0	71.8	57.6	42.7		
		MCWB	38.9	36.5	43.3	55.3	65.4	70.5	72.1	70.8	71.1	62.3	52.6	40.9		
	2%	DB	36.8	37.0	46.2	63.6	78.6	83.4	85.2	82.5	78.1	65.2	51.9	38.7		
		MCWB	35.3	33.7	38.3	49.7	61.6	68.2	71.0	68.9	67.9	56.6	48.3	36.9		
	5%	DB	33.7	34.3	42.5	57.9	72.8	79.2	81.8	79.5	74.2	60.2	48.1	36.1		
		MCWB	32.3	31.9	36.6	46.6	58.6	65.5	68.7	67.5	64.9	53.7	44.8	34.5		
	10%	DB	29.7	31.7	39.4	52.7	68.1	75.5	78.8	76.8	70.4	56.5	44.1	33.4		
		MCWB	28.0	29.7	34.6	43.5	56.1	63.1	66.9	66.4	62.8	51.9	41.3	32.0		
	0.4%	WB	39.4	38.3	45.9	59.5	69.1	74.2	76.4	74.3	72.8	64.1	54.3	41.5		
		MCDB	39.9	39.7	50.2	70.7	79.2	83.8	85.6	81.9	79.8	69.7	56.0	42.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	34.3	40.2	52.3	65.1	70.9	73.6	71.8	69.4	58.6	49.4	37.4		
		MCDB	36.4	35.9	44.6	59.2	73.9	80.2	81.2	78.5	76.1	63.0	51.1	38.4		
	5%	WB	32.5	32.3	37.5	47.9	61.7	68.2	71.3	70.0	66.9	55.2	45.1	34.6		
		MCDB	33.7	34.0	41.1	55.7	68.2	75.4	78.1	76.4	72.0	58.7	47.6	35.8		
	10%	WB	28.1	29.5	35.4	44.7	57.8	65.6	69.4	68.2	64.4	52.3	41.8	32.3		
		MCDB	29.6	31.8	38.7	51.2	65.9	72.5	76.0	74.6	69.5	55.7	43.7	33.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	17.3	16.4	16.8	19.5	18.7	18.3	18.1	18.0	14.2	11.8	12.7		
		MCDBR	15.3	15.4	18.1	26.8	27.7	24.6	22.1	21.6	22.2	20.6	17.3	12.3		
		MCWBR	14.9	13.4	12.7	16.4	14.4	12.6	11.0	11.1	13.0	13.5	14.6	12.1		
		MCDBR	15.0	14.6	15.8	23.3	23.3	21.0	19.5	18.6	19.4	17.9	16.0	12.2		
	5% WB	MCWBR	14.8	13.5	12.3	16.1	14.6	13.0	11.3	10.8	12.5	13.4	14.6	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.278	0.315	0.345	0.366	0.405	0.407	0.398	0.368	0.332	0.300	0.270		
	taud		2.318	2.248	2.256	2.382	2.401	2.284	2.293	2.325	2.399	2.506	2.504	2.356		
	Ebn at Noon		268	284	288	289	285	273	271	269	268	263	252	249		
	Edh at Noon		23	31	36	36	36	41	40	37	32	24	20	20		
(44) All-Sky Solar Radiation	RadAvg		448	730	1145	1433	1629	1778	1748	1530	1194	699	414	303		
	RadStd		64	85	105	138	172	130	161	71	96	56	53	25		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page